

Introduction

Exercises ISLR – Ch.11

Marcelo Previato Simoes Nº 2367070

27/10/2025



Conceptual

Exercises: 1(a-e), 4, 7, 9

Ex.1) Independence of Censored Events (a-e)

1. For each example, state whether or not the censoring mechanism is independent. Justify your answer.

(a) In a study of disease relapse, due to a careless research scientist, all patients whose phone numbers begin with the number “2” are lost to follow up.

Independent: no relation between phone number of occurrence of a disease

(b) In a study of longevity, a formatting error causes all patient ages that exceed 99 years to be lost (i.e. we know that those patients are more than 99 years old, but we do not know their exact ages).

Dependent: the censorship is directly related to the survival time (age)

(c) Hospital A conducts a study of longevity. However, very sick patients tend to be transferred to Hospital B, and are lost to follow up.

Dependent: the censorship is related to the gravity of the disease, which in its turn is related to the survival time

(d) In a study of unemployment duration, the people who find work earlier are less motivated to stay in touch with study investigators, and therefore are more likely to be lost to follow up.

Dependent: the censorship is related to the motivation, which in its turn is related to the timing of job finding

(e) In a study of pregnancy duration, women who deliver their babies pre-term are more likely to do so away from their usual hospital, and thus are more likely to be censored, relative to women who deliver full-term babies.

Dependent: the censorship is directly related to the deliver time

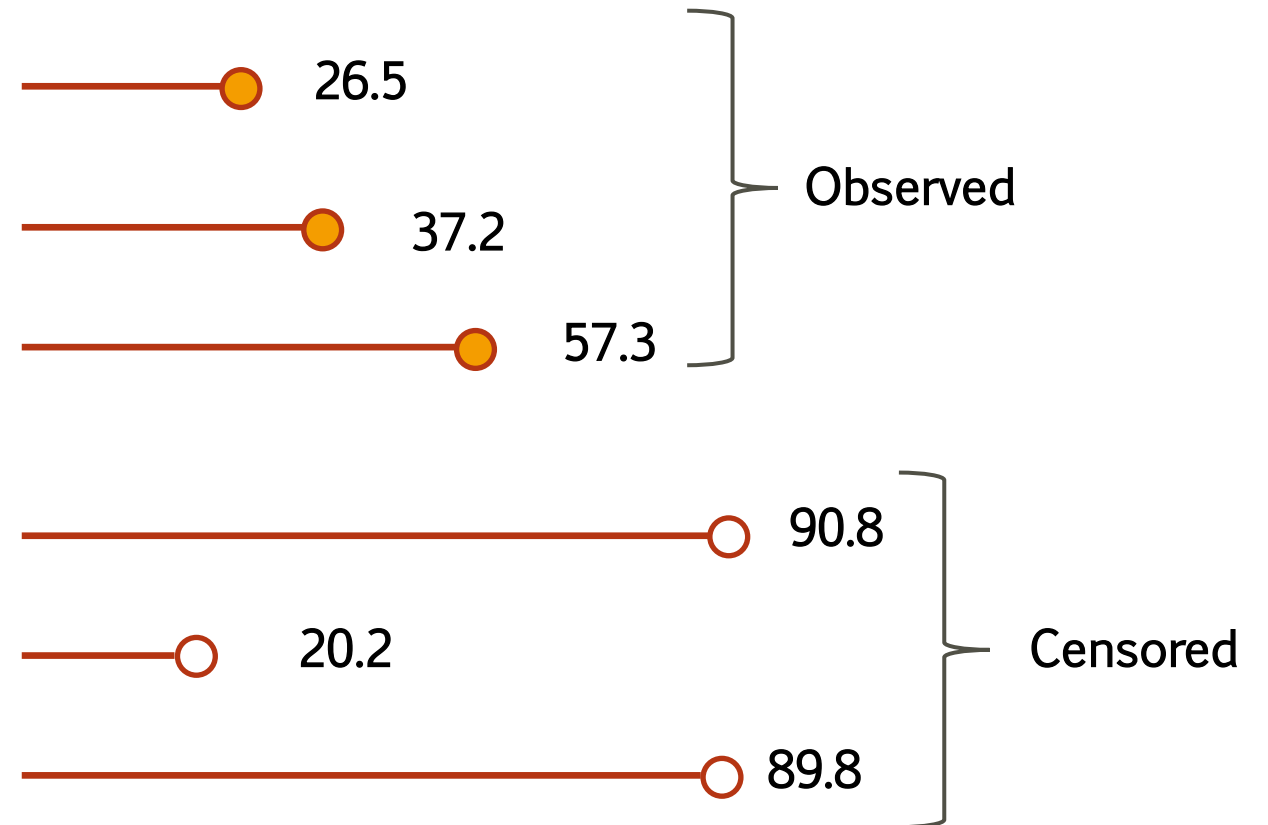
Ex.4) Kaplan-Meier Survival Curve

4. This problem makes use of the Kaplan-Meier survival curve displayed in Figure 11.9. The raw data that went into plotting this survival curve is given in Table 11.4. The covariate column of that table is not needed for this problem.

Observation (Y)	Censoring Indicator (δ)
26.5	1
37.2	1
57.3	1
90.8	0
20.2	0
89.8	0

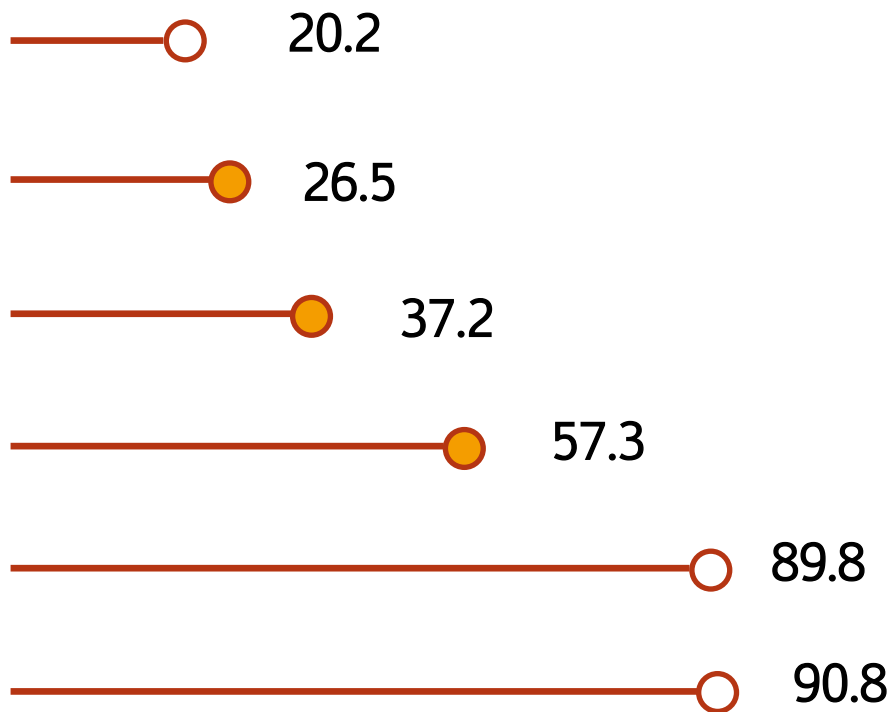
$$\delta = \begin{cases} 1 & \text{if } T \leq C \\ 0 & \text{if } T > C. \end{cases}$$

Thus, $\delta = 1$ if we observe the true survival time, and $\delta = 0$ if we instead observe the censoring time.



Ex.4) Kaplan-Meier Survival Curve

Ordering the events



Kaplan-Meier estimator $\hat{S}(d_k) = \prod_{j=1}^k \left(\frac{r_j - q_j}{r_j} \right)$

K	Time	r	q	(r-q)/r	S
1	d1 = 26.5	5	1	4/5	4/5
2	d2 = 37.2	4	1	3/4	3/5
3	d3 = 57.3	3	1	2/3	2/5

timing of
observed events

alive just
before dk

dead at
time dk

product of
all fractions
up to moment k

Ex.4) Kaplan-Meier Survival Curve (a)

(a) What is the estimated probability of survival past 50 days?

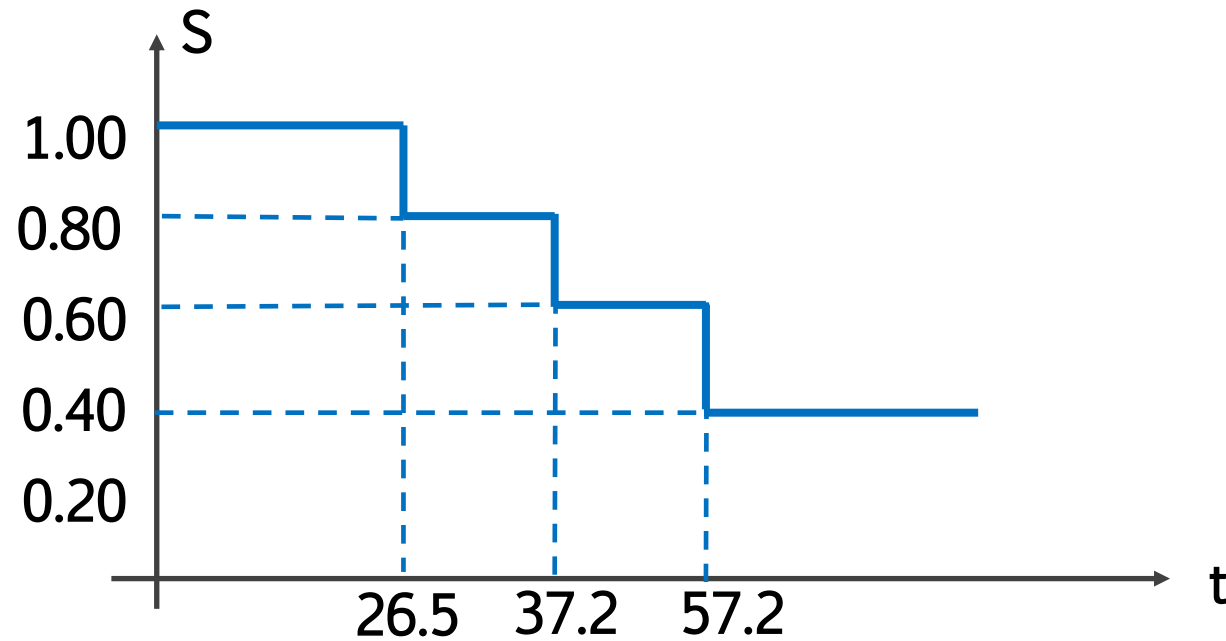
$$S(d_k) = \Pr(T > d_k) :$$

K	Time	r	q	(r-q)/r	S
1	d1 = 26.5	5	4	4/5	4/5
2	d2 = 37.2	4	3	3/4	3/5
3	d3 = 57.3	3	2	2/3	2/5

$$\Pr(T > 50) = S(50) = 3/5 = 0.60$$

Ex.4) Kaplan-Meier Survival Curve (b)

- (b) Write out an analytical expression for the estimated survival function. For instance, your answer might be something along the lines of



Obs: Only observed data
create steps

Ex.7) Log-Rank Test & Hypergeometric

7. In this problem, we will derive (11.5) and (11.6), which are needed for the construction of the log-rank test statistic (11.8). Recall the notation in Table 11.1.

	Group 1	Group 2	Total
Died	q_{1k}	q_{2k}	q_k
Survived	$r_{1k} - q_{1k}$	$r_{2k} - q_{2k}$	$r_k - q_k$
Total	r_{1k}	r_{2k}	r_k

TABLE 11.1. Among the set of patients at risk at time d_k , the number of patients who died and survived in each of two groups is reported.

Ex.7) Log-Rank Test & Hypergeometric (a)

- (a) Assume that there is no difference between the survival functions of the two groups. Then we can think of q_{1k} as the number of failures if we draw r_{1k} observations, without replacement, from a risk set of r_k observations that contains a total of q_k failures. Argue that q_{1k} follows a *hypergeometric distribution*. Write the parameters of this distribution in terms of r_{1k} , r_k , and q_k .

	Group 1	Group 2	Total
Died	q_{1k}	q_{2k}	q_k
Survived	$r_{1k} - q_{1k}$	$r_{2k} - q_{2k}$	$r_k - q_k$
Total	r_{1k}	r_{2k}	r_k

i) Under H_0 : same survival function → common probability of death (“success”) q_k/r_k in the population, with a total size r_k , total number of successes q_k e number of events q_{1k}

ii) Equivalent to a sampling without replacement → the total number of successes and failures are fixed for a given period, so that q_{1k} and q_{2k} are not independent

Therefore, q_{1k} follows a hypergeometric distribution:

$$q_{1K} \sim \text{Hypergeometric}(\text{total size} = r_k, \text{total successes} = q_k, \text{sample size} = r_{1k})$$

Ex.7) Log-Rank Test & Hypergeometric (b)

- (b) Given your previous answer, and the properties of the hypergeometric distribution, what are the mean and variance of q_{1k} ? Compare your answer to (11.5) and (11.6).

$$\Pr(X = x) = \frac{\overbrace{\binom{K}{x}}^{\text{combination of successes}} \overbrace{\binom{N-K}{n-x}}^{\text{combination of failures}}}{\underbrace{\binom{N}{n}}_{\text{total of combinations}}} \quad \Rightarrow \quad \Pr(q_{1x}) = \frac{\binom{q_k}{q_{1x}} \binom{r_k - q_k}{r_{1k} - q_{1x}}}{\binom{r_k}{r_{1k}}}$$

Using the properties of the hypergeometric distribution:

$$\mathbb{E}(q_{1x}) = r_{1k} \frac{q_k}{r_k}$$

$$\text{Var}(q_{1x}) = r_{1k} \frac{q_k}{r_k} \left(1 - \frac{q_k}{r_k}\right) \frac{r_k - r_{1k}}{r_k - 1}$$

$$\mu_k = \frac{r_{1k}}{r_k} q_k \quad \checkmark$$

$$\sum_{k=1}^K \text{Var}(q_{1k}) = \sum_{k=1}^K \frac{q_k (r_{1k}/r_k) (1 - r_{1k}/r_k) (r_k - q_k)}{r_k - 1} \quad \checkmark$$

Ex.9) Survival Time ~ Exponential (a)

9. In this exercise, we will explore the consequences of assuming that the survival times follow an exponential distribution.

(a) Suppose that a survival time follows an $\text{Exp}(\lambda)$ distribution, so that its density function is $f(t) = \lambda \exp(-\lambda t)$. Using the relationships provided in Exercise 8, show that $S(t) = \exp(-\lambda t)$.

Given the probability density function of the exponential distribution, it is possible to get the accumulated function and the survival function:

$$f(t) = \lambda e^{-\lambda t} \Rightarrow F(t) = \int_0^t \lambda e^{-\lambda x} dx = -e^{-\lambda t} + 1$$

$$F(t) = 1 - S(t) \Rightarrow S(t) = 1 - F(t) = 1 - (-e^{-\lambda t} + 1) = e^{-\lambda t}$$

$$S(t) = e^{-\lambda t}$$

Ex.9) Survival Time ~ Exponential (b)

- (b) Now suppose that each of n independent survival times follows an $\text{Exp}(\lambda)$ distribution. Write out an expression for the likelihood function (11.13).

$$L_i = \begin{cases} f(y_i) & \text{if the } i\text{th observation is not censored} \\ S(y_i) & \text{if the } i\text{th observation is censored} \end{cases}$$
$$= f(y_i)^{\delta_i} S(y_i)^{1-\delta_i}.$$

$$L = \prod_{i=1}^n f(y_i)^{\delta_i} S(y_i)^{1-\delta_i}$$

where the exponents are the indicator on censorship (1 = observed, 0 = censored)

$$L(t) = \prod_{i=1}^n (\lambda e^{-\lambda t_i})^{\delta_i} (e^{-\lambda t_i})^{1-\delta_i} = \lambda^{\sum_{i=1}^n \delta_i} \cdot e^{-\lambda (\sum_{i=1}^n t_i)}$$

Sum of observed events
(not censored)

Sum of all times, observed
and censored

Ex.9) Survival Time ~ Exponential (c)

(c) Show that the maximum likelihood estimator for λ is

$$\hat{\lambda} = \sum_{i=1}^n \delta_i / \sum_{i=1}^n y_i.$$

Calculating the log-likelihood and getting the first derivative equal to 0, it is possible to find the MLE:

$$L(t) = \lambda^{\sum_{i=1}^n \delta_i} \cdot e^{-\lambda(\sum_{i=1}^n t_i)}$$

$$l(t) = \sum_{i=1}^n \delta_i \cdot \ln(\lambda) - \sum_{i=1}^n t_i \cdot \lambda$$

$$\frac{\partial l(t)}{\partial \lambda} = 0 \Rightarrow \frac{\sum_{i=1}^n \delta_i}{\lambda} - \sum_{i=1}^n t_i = 0$$

$$\lambda_{MLE} = \frac{\sum_{i=1}^n \delta_i}{\sum_{i=1}^n t_i}$$

Sum of observed events
(not censored)

Sum of all times, observed
and censored

Ex.9) Survival Time ~ Exponential (d)

(d) Use your answer to (c) to derive an estimator of the mean survival time.

Hint: For (d), recall that the mean of an $\text{Exp}(\lambda)$ random variable is $1/\lambda$.

By the invariance property of MLE:

$$\left(\frac{1}{\lambda}\right)_{MLE} = \frac{1}{\lambda_{MLE}} = \frac{\sum_{i=1}^n t_i}{\sum_{i=1}^n \delta_i}$$

Sum of all times, observed and censored

Sum of observed events (not censored)